

Experiment No. 15

Title

Decision Tree Using Artificial Intelligence

Aim

To develop a Python program to implement the Decision Tree algorithm for classification.

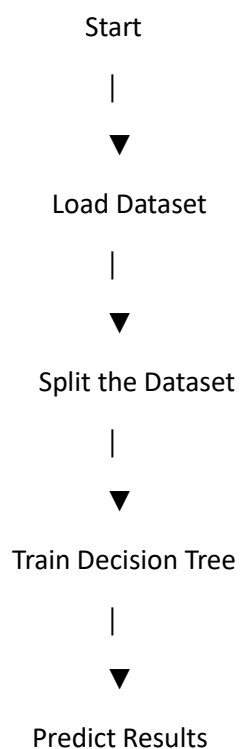
Objective

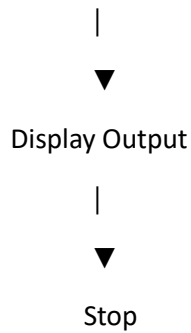
The objective of this experiment is to understand the Decision Tree algorithm and use it to classify data based on its features.

Algorithm

- Step 1:** Import the required libraries.
 - Step 2:** Load the dataset.
 - Step 3:** Split the dataset into training and testing sets.
 - Step 4:** Train the Decision Tree classifier.
 - Step 5:** Predict the test data.
 - Step 6:** Display the predicted output.
 - Step 7:** Stop.
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Flowchart





Python Program

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier


# Load dataset

iris = load_iris()

X = iris.data

y = iris.target


# Split dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)


# Train Decision Tree

model = DecisionTreeClassifier()

model.fit(X_train, y_train)


# Predict

prediction = model.predict(X_test)


print("Predicted Values:")

print(prediction)
```

Sample Output

Predicted Values:

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 2 2 2 2 2 0 0 0 0 1 0 0 2 1  
0 0 0 2 1 1 0 0]
```

Result

The Decision Tree classifier successfully classified the given dataset.

Conclusion

Thus, the Decision Tree algorithm was successfully implemented. It effectively classifies data by constructing a tree based on feature values and is widely used for classification problems in Artificial Intelligence and Machine Learning.